

MELAKU ALEHEGN

Full Stack AI / Machine Learning Engineer

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Summary

Full Stack AI and Machine Learning Engineer with production experience in demand forecasting, multi-tenant ETL/ELT pipelines (Airflow, DBT, Snowflake), and LLM-based systems. Strong focus on end-to-end pipelines, data quality, feature engineering, and integrating models into real-world products. Comfortable working across data, backend, and infrastructure layers to deliver reliable, scalable solutions.

Experience

Wicked Analytics, *Machine Learning / Data Engineer* 02/2026 – Present | Remote

- Designed and deployed a production ML order-forecasting system serving 11 retail-supplier tenants selling into Walmart, producing per-item, per-horizon 16-week sales and order forecasts that drive weekly replenishment decisions. Lifted forecast accuracy from 81 percent (raw gradient-boosted baseline) to 88.6 percent on the client-facing view through a stacked correction pipeline.
- Architected multi-tenant ETL/ELT pipelines in Airflow, ingesting per-client sales, forecasts, supply plans, and inventory into Snowflake via Azure Blob and transforming them with incremental DBT models; exposed current-run and historical views powering instant Domo KPIs and automated Excel deliverables.
- Developed a full-stack demand planning web application (React, FastAPI, PostgreSQL) for planners: interactive item list, editable per-week order plans, projected-inventory cascade computed on the fly, and multi-vendor support, replacing legacy spreadsheet-only workflows.

Arifpay, *Data Scientist / Engineer* 01/2025 – 05/2026 | Addis Ababa, Ethiopia

- Built and deployed machine learning models for payment fraud detection, working with high-volume transactional data and severe class imbalance
- Designed behavioral and temporal feature pipelines to improve fraud signal quality, contributing to a 15% reduction in fraud incidents
- Developed a merchant credit scoring system using FastAPI, improving lending decision accuracy and increasing approval confidence by around 40%
- Implemented continuous learning pipelines using MLflow and Celery to retrain and monitor models in production
- Built and maintained Airflow DAGs for ML workflows, enabling reliable scheduling and monitoring and helping prevent an estimated \$100K in potential annual losses
- Designed Metabase dashboards for executive-level KPIs (revenue, churn, acquisition, efficiency), enabling data-driven decision-making across the business
- Built Retrieval-Augmented Generation (RAG) systems over internal documents and operational data to support internal analytics and decision support workflows, focusing on reliability and real-world usability

Openstack, *Software Developer Intern* 12/2023 – 04/2024 | Remote

- Enhanced the UI workflow of the OpenStack Manila project using Django views, streamlining the share network creation process and improving usability.
- Increased test reliability by achieving 90% unit test coverage and introducing pre-commit and tox configurations to standardize development workflows.

- Collaborated with OpenStack core reviewers to develop, refine, and merge patches within release cycles in a large-scale open-source environment.

Nedamco Africa, *Cloud Consultant*

12/2023 – 03/2024 | Addis Ababa, Ethiopia

- Consulted clients across agritech, hospitality, and pharmaceutical industries on cloud adoption strategies using AWS and Azure platforms.
- Assisted in designing scalable cloud architectures and guided clients on selecting services for compute, storage, networking, monitoring, and cost optimization.
- Supported cloud migration and deployment planning while working with AWS services including EC2, S3, IAM, RDS, and CloudWatch.
- Participated in stakeholder discussions and prepared technical recommendations, architecture documentation, and implementation guidance aligned with business requirements.

Projects

Pallet *🔗*, *Grounded AI agent for inventory decision support*

Stack: LangGraph, Gemini API, FastAPI, LightGBM, DuckDB, MLflow

A LangGraph + Gemini agent for demand and inventory decisions that runs forecasting and simulation tools on demand, compares what-if scenarios, and answers in plain language - with a deterministic grounding check that verifies every number against the tool output that produced it, so it can't hallucinate figures. Backed by a LightGBM quantile forecaster and a base-stock policy simulation, with an MLflow model registry, Langfuse tracing, and tests in CI.

Verba *🔗*, *Full Stack Multi-tenant RAG platform*

Stack: FastAPI, Qdrant, Celery, React, Gemini API

Async ingestion (parse→chunk→embed), retrieval with MMR reranking + hybrid (keyword+vector) search, conversational query rewriting, token-streamed answers with source citations, and a strict grounding threshold that refuses to hallucinate. Hardened with OAuth, per-IP rate limiting, structured logging, readiness probes, and a 140+ test suite with CI.

Chimera *🔗*, *Autonomous multi-agent system*

Stack: Java 21, Gemini API, PostgreSQL, MCP

Manager/Worker/Judge agent architecture with LLM-backed reasoning, event-sourced memory, a custom Model Context Protocol client written from scratch, and autonomous publishing to social media.

Education

Addis Ababa Science & Technology University

09/2016 – 09/2021 | Addis Ababa, Ethiopia

BSc in Electrical and Computer Engineering

Skills

Applied AI / ML (Python • Pandas • NumPy • Polars • Scikit-learn • TensorFlow • Feature Engineering • Model Evaluation) | **Data & MLOps** (Apache Airflow • dbt • Dimensional Modeling • MLflow • DuckDB • Celery • Kafka • Snowflake • ETL/ELT • Data Warehousing • Kafka) | **Backend & Deployment** (FastAPI • Flask • Docker • SQL • PostgreSQL • REST APIs • Redis • Node.js • CI/CD) | **LLMs & Search** (LangChain • LangGraph • Langfuse • Vector Databases (ChromaDB, Qdrant, PgVector) • MCP • Multi-Agent Systems) | **Cloud & Analytics** (Azure • AWS • GCP • Metabase)